Language Models, Embeddings, and Sentiment Representations

DSA4213 Assignment 1 | IMDb | Submission date: 16 October 2026

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Research question and main findings

Which model predicts the next token best on a small common corpus, how do frozen embeddings change an LSTM, and do language-model representations transfer to sentiment classification? Splits are fixed and deduplicated. Hyperparameters use validation data, but supplementary experiment design was informed by original test results (p. 6).

Prediction. In Part I, lower test PPL ranks LSTM (170.46), RNN (182.34), Transformer (205.10), then trigram (220.60). Grid-tuned trigram reaches 216.02; Transformer cap 40 reaches 204.28 (Table 3). This ordering is specific to this corpus, model scale and training setup.

Embeddings. Three-seed mean +/- sample SD test PPL is random trainable: 171.89 +/- 1.31; frozen Skip-gram: 170.94 +/- 0.92; frozen GloVe: 173.80 +/- 2.40. Random frozen reaches 208.90 (one seed). Frozen pretrained inputs are broadly comparable to trainable inputs, without a clear overall improvement (Table 5).

Transfer. TF-IDF plus logistic regression reaches 86.3% test accuracy, above both frozen LM representations. Transformer reaches 70.7% vs 69.1% for LSTM, but its paired accuracy difference has a 95% interval of [-1.4, 4.7] percentage points and exact McNemar p=0.331. This does not establish a reliable downstream ordering.

1. Data and evaluation contract

IMDb is loaded as stanfordnlp/imdb [1]. Official test reviews are deduplicated first. Training reviews matching any normalized official-test text are removed; other duplicates keep their first occurrence. Keys lowercase text and collapse whitespace. With split seed 42, a stratified 6,000-review training-pool sample is divided into 5,000 training and 1,000 validation reviews; 1,000 test reviews are sampled from the separate official test pool.

Table 1. Data splits and token statistics.

Split	Reviews	Negative / positive	Tokens	UNK
Train	5,000	2,489 / 2,511	1,353,941	1.882%
Validation	1,000	498 / 502	270,132	2.813%
Test	1,000	498 / 502	276,278	3.304%

Table 1 summarizes the split. HTML is unescaped, line breaks removed and text lowercased. A regular expression retains words with internal apostrophes, numbers and punctuation. Train-only counts define 19,996 ordinary tokens (minimum count 2), sorted by decreasing frequency with alphabetical ties, plus PAD/UNK/BOS/EOS. Unseen words map to UNK.

Every model predicts from preceding context only. Cross-entropy (CE) is total negative natural-log likelihood / valid targets; PPL=exp(CE). PAD is excluded, BOS is never a target, and each review contributes one EOS: 271,132 validation and 277,278 test targets. Comparisons are conditional on this vocabulary and tokenization. The 1.35M-token training corpus limits what these models can learn.

Task-specific evaluation. Language modeling uses CE/PPL; sentiment classification uses accuracy and macro-F1. Generated text also requires coherence assessment. These criteria answer different questions and cannot substitute for one another.

2. Part I: prediction quality and cost

Table 2. Model architectures and gradient-trained parameter counts.

Model	Architecture / context	Gradient-trained parameters
Trigram	Count estimation; two preceding tokens	0 (count-based)
RNN	One tanh layer; hidden size 256	7,231,648
LSTM [5]	One layer; hidden size 256	7,506,592
Transformer [6]	2 causal layers; 4 heads; d=100; FFN=400	4,275,600

Table 2 describes each architecture. Trigram training estimates count tables and probabilities from the training corpus. It learns from data, but has no neural weights optimized by gradients; the zero above excludes its learned count-based state.

Neural inputs use trainable 100-dimensional embeddings and untied output projections. BOS+tokens and tokens+EOS are sliced into aligned chunks of at most 128 positions. All review targets are retained. Recurrent states and Transformer positions reset at chunk boundaries. Transformer uses learned positions, GELU, pre-layer normalization and dropout 0.1.

The original runs use Adam [7], learning rate 0.001, batch size 16, gradient clipping at 1, seed 42 and a 20-epoch cap. Three epochs without validation CE improving by more than 0.0001 from the last significant improvement trigger stopping. Every strictly lower validation CE saves a checkpoint; testing restores the absolute minimum. The 40-epoch Transformer repeats the same setup from scratch with only the cap changed.

Table 3. Language-model prediction and computational cost.

Model / run	Test CE	Test PPL	Train s	Size MB	Score k/s
Trigram	5.3964	220.60	3.0	17.57	605.6
RNN	5.2059	182.34	159.7	29.27	257.5
LSTM	5.1385	170.46	236.9	30.37	174.0
Transformer	5.3235	205.10	412.2	17.45	206.6
Trigram grid	5.3754	216.02	3.0	17.57	572.5
Transformer cap 40	5.3195	204.28	464.6	17.45	303.9

Table 3 notes. Train excludes validation, I/O and hyperparameter search. Trigram grid reuses the 3.0 s count fit; its size is verified after updating weights. Size is serialized bytes / 1,000,000. Score is thousands of test targets/s: first 10,000 targets, log loss included, one warm-up, median of three runs; CUDA synchronized. All timings are from the 15 October rerun with Python 3.12.13. Neural runs use RTX 3060 Laptop GPU; trigram uses CPU (Intel64 Family 6 Model 154). The same Transformer architecture scores 206.6 vs 303.9 k/s across runs (47% spread): a rough indicator of timing noise, not a formal noise estimate. This short workload and uncontrolled runtime state do not support a speed ranking.

Interpolation and budget sensitivity

Trigram unigram probabilities use add-0.1 smoothing. Bigram interpolation weight w2 and then trigram weight w3 give effective weights ((1-w3)(1-w2), (1-w3)w2, w3) when both contexts occur. Unseen contexts retain the lower-order distribution. A 400-pair validation grid, w2,w3 in 0,0.05,...,0.95, selects (0.6, 0.25), improving on the original three-pair search. This is interpolated maximum likelihood, not Kneser-Ney smoothing.

The extra-budget Transformer selects epoch 22 after 25 epochs (early stopping), compared with epoch 19 in the original run. The 0.40% test-PPL reduction leaves the original Part I ordering unchanged. Early stopping is a validation criterion, not proof of a global optimum; larger models or a different learning-rate schedule remain untested.

Training time per epoch, time to the selected checkpoint and autoregressive generation speed are reported in Appendix B. Total run time includes patience epochs after the best checkpoint, so it should not be interpreted as the cost of reaching that quality.

3. Part I: learning and generated text

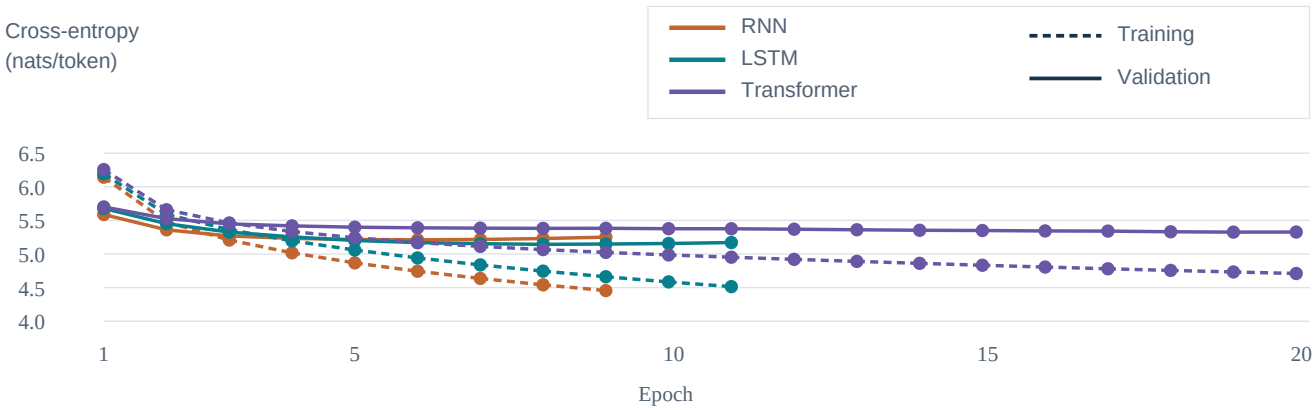


Figure 1. Seed-42 training (dashed) and validation (solid) cross-entropy.

RNN/LSTM validation CE rises while training CE falls after the selected epochs 6/8, consistent with overfitting. Transformer selects epoch 19 and reaches the 20-epoch cap; this alone does not establish convergence. The 40-epoch check improves PPL by only 0.40%.

Matched decoding over more prompts

Generation uses the baseline seed-42 neural checkpoints and trigram interpolation weights (w2,w3)=(0.5,0.4), not the grid-selected (0.6,0.25). Six models each produce three samples for five prompts under temperature-1 full sampling and top-k=40: 180 continuations, capped at 40 new tokens and stopped at EOS. PAD/BOS are excluded; UNK is retained. Independent model/prompt/sample streams are reused across decoding policies.

Table 4. Generation diversity, repetition and unknown-token rates.

Model	Full D2	Top-40 D2	Full repeat-3	Top-40 repeat-3	Top-40 UNK
Trigram	0.966	0.808	0.000	0.002	0.043
RNN	0.972	0.871	0.000	0.000	0.018
LSTM	0.952	0.816	0.005	0.000	0.026
Transformer	0.925	0.795	0.000	0.009	0.019
Frozen Skip-gram	0.935	0.809	0.005	0.005	0.028
Frozen GloVe	0.937	0.792	0.000	0.004	0.017

In Table 4, D2 is unique bigrams / all bigrams, pooled without crossing sample boundaries. Repeat-3 is within-continuation repeated trigram occurrences / all trigram occurrences. UNK is the generated-token share. These describe output diversity, repetition and unknown tokens; none establishes semantic coherence. High trigram diversity can coexist with incoherent sentences. Length and decoding affect these measures; full results are in generation.json.

Concrete continuations: top-k=40, sample 0

Trigram: . it starts off the set , the film has received . . .) i mean . and <unk> , a " the man , and in this case both . and it's <unk> <unk> with <unk> <unk> the

LSTM: also one of the funniest thing is about the plot that the action is a great story . the story - - that , it's still a lot of the movie that she doesn't have . the acting is in

Transformer: just that one of a <unk> but the characters was great and stupid . the best character i thought that they were on in my life . the first place where the time at the last 5 . it's not

In this one-prompt, sample-0 illustration, trigram fragments into punctuation/UNK; LSTM repeats the story without development; Transformer combines "great and stupid" with unclear referents. RNN starts with praise but drifts into broken syntax. Skip-gram loses its subject after "a very different, but"; GloVe shifts from a TV show to an unclear viewing account. These are readings of examples, not model-wide judgments (Appendix C).

Only 15 continuations per model/decoder support Table 4. A repeat-3 rate of 0.005 is about three repeated occurrences among roughly 600 trigrams. Small D2/repeat-3 differences cannot rank quality; larger D2 differences still describe lexical diversity, not coherence.

4. Part II: embeddings and seed variability

The Part I LSTM is reused as the random-trainable baseline. Own vectors use Skip-gram negative sampling [3,4]: dimension 100, window 5, five negatives, min count 2, subsampling 0.001, five epochs, one worker, seed 42. Public vectors are GloVe 6B 100d [2]. Both pretrained tables are entirely frozen, including missing and special rows; remaining LSTM parameters train from scratch. Trainable parameters decrease from 7,506,592 to 5,506,592.

All variants use the same splits, architecture, Adam settings and stopping policy. Seeds 42/43/44 vary LM initialization and batch order. Pretrained vectors and Normal(0,1) missing rows remain fixed at seed 42, with zero PAD. Mean and sample standard deviation below therefore measure optimization variability conditional on these embeddings, not variability from retraining Skip-gram or resampling data.

Table 5. Part II test perplexity across optimization seeds.

LSTM input	Seed 42 PPL	Seed 43 PPL	Seed 44 PPL	Mean +/- SD	Train to best s (seed 42)
Random trainable	170.46	173.03	172.18	171.89 +/- 1.31	171.8
Frozen Skip-gram	170.02	170.93	171.85	170.94 +/- 0.92	278.5
Frozen GloVe	175.42	174.94	171.05	173.80 +/- 2.40	377.3

Time is seed-42 training through the best checkpoint, excluding validation/pretraining. The LSTM architecture is identical: 21.5/28.2/37.8 s per epoch differ by 75%, indicating uncontrolled runtime variation, not an identified cost of freezing. Times cannot rank embedding efficiency. Three seeds and one split do not establish general superiority.

A seed-42 random-frozen control gives test PPL 208.90, versus 170.46 with identical initialization but trainable inputs: freezing harms prediction here. Both pretrained-frozen tables improve on random-frozen at seed 42, recovering much of this loss without input trainability. Domain, objective and scale still differ; this control is one seed.

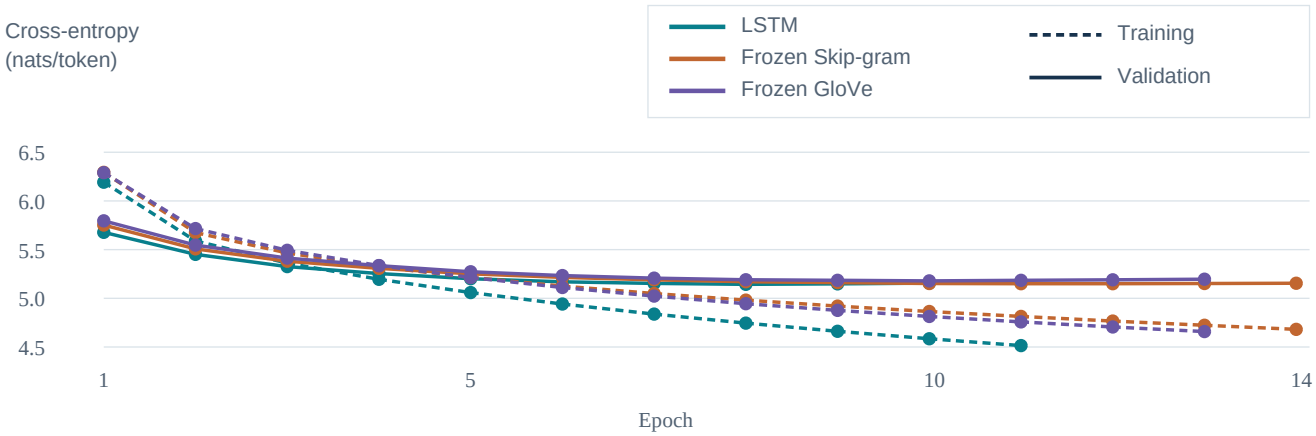


Figure 2. Seed-42 learning curves: best epochs 8/11/10, stopping at 11/14/13. Freezing delays the optimum here but does not prevent overfitting. Table 5 supplies the evidence about seed variability.

Coverage, scale and cost

Skip-gram covers all 19,996 ordinary LM words and 100% of their training occurrences. GloVe covers 19,042/19,996 words (95.23%) and 98.38% of occurrences restricted to ordinary LM words. It leaves 954 random ordinary rows. Relative to all original training tokens (including out-of-vocabulary words), coverage is 98.12%/96.53%. These denominators answer different questions.

Mean ordinary-row L2 norms: random initial 9.98, Skip-gram 1.48, GloVe 5.20. Unequal scale is a remaining confound; no normalized-table ablation was performed. Differences cannot be attributed solely to pretraining quality.

Skip-gram uses only the same 1.35M-token training corpus: it adds no data yet roughly matches trainable inputs. GloVe adds 6B external Wikipedia/Gigaword tokens [2] without winning here; domain mismatch with reviews, 954 random ordinary-word rows and unequal norms are possible explanations, none isolated by this experiment.

Skip-gram pretraining costs 45.47 s separately. A local GloVe parsing pass costs 0.81 s; historical download time was not recorded, and public pretraining cost is excluded. Per-epoch LM costs are in Appendix B.

5. Part III: sentiment representations

The baseline Part I checkpoints are frozen. Contextual states before the vocabulary projection are averaged over ordinary and UNK token positions, excluding PAD/BOS/EOS. Chunk states are weighted by token count, retaining the causal/reset policy. LSTM and Transformer features have 256 and 100 dimensions.

Logistic regression (L-BFGS, max 2,000 iterations) selects C in {0.001,0.01,0.1,1,10} by validation macro-F1. Dense-feature scaling and the 100d PCA control are train-fit. TF-IDF uses the same tokenizer, 1-2 grams, min_df=2, at most 50,000 features, sublinear TF and L2 normalization; vocabulary/IDF are train-fit. No external vectors enter this baseline.

Table 6. Sentiment classification and selected regularization.

Representation	Selected C	Val. macro-F1	Test accuracy	95% accuracy CI	Test macro-F1
LSTM mean, 256d	0.1	0.7709	69.1%	[66.2, 72.0]	0.6910
Transformer mean, 100d	0.01	0.7537	70.7%	[67.9, 73.6]	0.7068
LSTM + PCA, 100d	0.001	0.7326	67.0%	[64.0, 69.9]	0.6698
TF-IDF, 50,000d	10	0.8900	86.3%	[84.2, 88.4]	0.8630

Table 6 intervals use 10,000 paired test-review bootstrap resamples (seed 4213), with fixed models; they exclude retraining variability. Majority accuracy is 50.2%. LSTM (C=0.1) and Transformer (C=0.01) are now interior selections. PCA (C=0.001) and TF-IDF (C=10) remain at grid boundaries. Values beyond this grid remain untested, so optimal regularization is not established.

Transformer makes 293 errors vs 309 for LSTM. Their discordant counts are 111 (LSTM only correct) and 127 (Transformer only correct). Exact two-sided McNemar p=0.331; the paired macro-F1 difference interval is [-0.0142, 0.0465]. Both difference intervals include zero. Validation favors LSTM (0.7709 vs 0.7537), so the test ordering is not a stable architecture ranking.

Table 6 shows a stronger TF-IDF baseline. PCA does not close the gap (67.0% accuracy); it controls dimension, not architecture. Validation-to-test macro-F1 drops are 0.0799 for LSTM, 0.0469 for Transformer and 0.0270 for TF-IDF. The TF-IDF drop shows that LM exposure cannot be the sole explanation; sampling differences and validation-based selection may contribute.

My reading of Appendix D: #922 and #605 mix praise with rejection of execution, while #254 uses positive words in a negative judgment and #447 explicitly praises a film classified negative. The full #965 review mixes pacing criticism with a positive ending; a truncated excerpt can hide that conclusion. #987 appears positive despite its negative label, suggesting a possible label inconsistency, not a verified error in annotation.

Error groups and unverified split hypothesis

Table 7. Descriptive accuracy by review length and UNK share.

Review group	n	LSTM accuracy	Transformer accuracy
length<=128	158	67.1%	70.3%
length>256	362	68.5%	69.9%
UNK>2%	625	69.8%	70.6%

Table 7 groups overlap. The short-review group has only n=158: the 3.16-point Transformer advantage is five reviews; paired exact McNemar p=0.533 (18 vs 23 discordant cases), compatible with sampling noise. Appendix D contains randomly sampled errors.

Maas et al. [8] constructed official train/test sets with disjoint movies. Our train/validation subsets both come from the official train pool, so they may share movies. HF plain_text exposes text and label, not movie IDs [1]; overlap is unverified here. Both validation and test reviews were unseen by the LM: exposure to training reviews alone cannot explain their contrast. Representation overfitting could contribute to the LM-specific excess drop over TF-IDF, but this is an untested hypothesis, not a causal decomposition.

6. Interpretation and reproducibility

What the comparisons support

Prediction quality is conditional on the experiment. LSTM achieves lower CE/PPL than Transformer on this small corpus at the tested model sizes and training settings. The trigram grid and extra-budget Transformer probe two comparison weaknesses. Possible explanations are not isolated: Transformer/LSTM have 4.28M/7.51M parameters, including 2.02M/5.14M in their output layers. Both use chunks of at most 128 positions, versus two tokens for trigram. Completed epochs and training times also differ (Table 3). These factors may contribute; this is not a universal architecture ranking.

Frozen pretraining roughly matches trainable inputs. The lowest three-seed mean test PPL belongs to frozen Skip-gram. Three seeds are a small estimate of optimization variability, with fixed embeddings. The random-frozen control helps separate freezing from vector source; unmatched norms, external GloVe data and coverage still limit attribution. A tiny single-seed improvement should not be used as evidence of general superiority.

Language modeling and sentiment need different evidence. Table 6 shows a stronger supervised lexical baseline than either frozen LM feature set. LSTM has lower LM perplexity (170 vs 205). Transformer is nominally higher on downstream test accuracy, but its paired difference CI includes zero; validation favors LSTM. There is no evidence establishing either agreement or reversal of the two rankings. Random error examples in Appendix D illustrate failures but do not identify their internal causes.

Remaining limits

Training uses only 1.35M tokens and one split. Multi-seed runs hold pretrained tables fixed; the architecture comparison, extra-budget run and random-frozen control are single-seed. Generation uses five prompts, three samples each and no human fluency annotation. Diversity does not measure meaning. Exact deduplication misses near-duplicates. Test bootstrap intervals omit training variability; error groups overlap. Supplementary experiments were designed after inspecting original test results. Validation-based hyperparameter selection does not undo that influence on the research process: the test set is no longer untouched. Supplementary findings are exploratory evidence; confirmation requires fresh held-out data.

Reproduction package

Assignment.ipynb is the complete execution path: restart the kernel and run all cells after preparing dependencies and GloVe. Original and extended experiments, result checks and provenance are integrated in topic order, with no automatic reuse of old training JSON. Section notes map inputs to output files. scripts/build_report.py only formats the PDF. The complete notebook was rerun on 15 October 2026, 00:16-01:43 Singapore time, and reached the final checks. On 16 October, only classifiers and their statistics were rerun from saved features with the expanded C grid; run_provenance.json records this partial update separately. LM results, generation and timings remain from the full run.

The experiments used Python 3.12.13 and package versions recorded in requirements.txt and results/supplementary/environment.json. Neural experiments use PyTorch 2.14.0+cu130 on an RTX 3060 Laptop GPU; CPU execution is supported but slower and timings will differ. IMDb and GloVe 6B 100d are obtained from the cited sources; the expected vector path is data/glove.6B/glove.6B.100d.txt. Split and vocabulary hashes are retained for checking reproduction. Exact numerical identity across devices is not guaranteed.

Sources, attribution and tool use

The implementation uses PyTorch, gensim and scikit-learn. Model, algorithm and dataset references [1-8] appear after the appendices. No course notebook was adapted. OpenAI Codex assisted with code, supplementary experiments, statistical calculations, report editing and PDF preparation. I am responsible for the experimental design and conclusions and have checked them.

The code archive includes the notebook, PDF builder, report, README, requirements, numeric results and saved test predictions. It excludes raw review splits, token-ID files, GloVe vectors, model checkpoints and classifier feature/model files. GloVe 6B 100d must be downloaded; the notebook regenerates other excluded artifacts. The full local working folder also retains these artifacts. README documents the same packaging distinction.

Appendix A. Additional training evidence

Table A1. Validation checkpoints and seed-level perplexities.

Variant / seed	Best epoch	Test PPL	Validation PPL
random trainable / 42	8	170.457	171.309
random trainable / 43	9	173.032	173.111
random trainable / 44	9	172.176	173.058
own frozen / 42	11	170.022	172.419
own frozen / 43	13	170.933	172.997
own frozen / 44	12	171.853	173.930
glove frozen / 42	10	175.421	177.494
glove frozen / 43	11	174.936	176.159
glove frozen / 44	10	171.049	172.726

Extra-budget Transformer learning curve

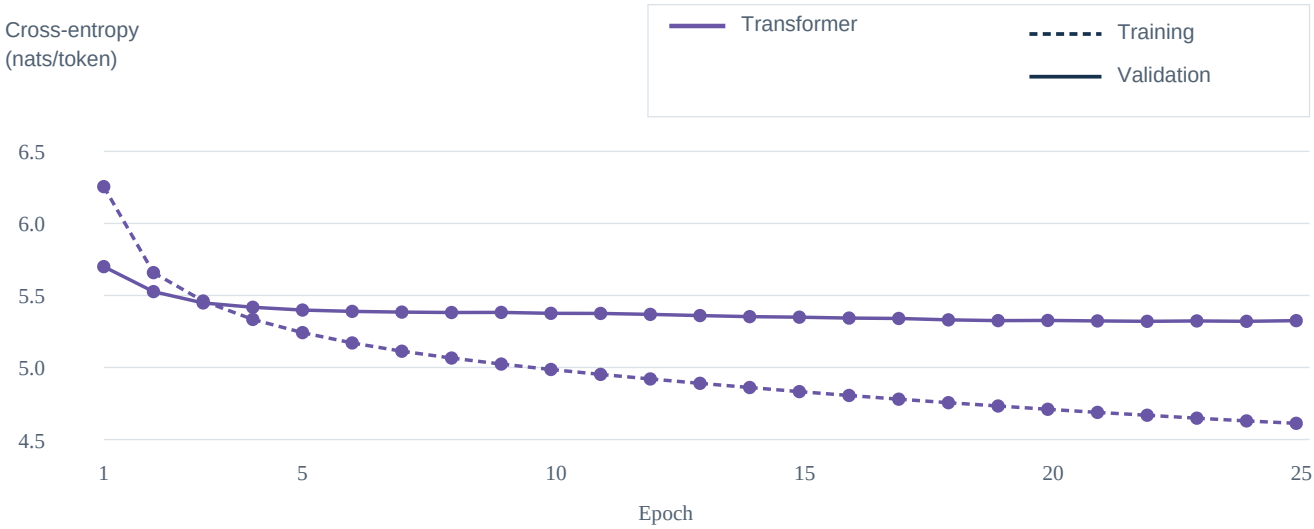


Figure A1. Extra-budget Transformer training and validation CE. Table A1 reports seed-level results. The same-seed fresh run executes 25 epochs; selected epoch 22, stop reason early_stopping. Test CE 5.319515, PPL 204.2847. The optimizer and learning rate are unchanged. Its history is saved separately from the original 20-epoch experiment.

Random-frozen control: best epoch 9, 12 epochs executed, test PPL 208.9023. The complete input table is frozen, including special rows.

Appendix B. Timing and embedding diagnostics

Table B1. Training time per epoch and to the best checkpoint.

Model / embedding (seed 42)	Train s / epoch	Train s to best	Train + validation s to best
RNN	17.75	106.31	113.76
LSTM	21.54	171.79	184.08
Transformer	20.61	389.68	414.56
Frozen Skip-gram	28.20	278.49	301.56
Frozen GloVe	37.78	377.28	413.67

Table B1 measures time to best across epochs 1 through the selected epoch, and excludes checkpoint I/O and preprocessing. Per-epoch means cover all executed epochs. These timings come from the complete 15 October rerun. Sequential measurements can reflect changing runtime conditions; they do not isolate architectural efficiency.

Table B2. Autoregressive generation throughput.

Model	Full sample tokens/s	Top-40 tokens/s	Full generated tokens	Top-40 generated tokens
Trigram	53.8	50.6	537.0	583.0
RNN	2175.6	1431.2	510.0	495.0
LSTM	2074.7	1269.8	600.0	576.0
Transformer	686.7	580.6	600.0	591.0
Frozen Skip-gram	2009.2	1404.7	600.0	600.0
Frozen GloVe	2013.7	1399.4	600.0	596.0

Table B2 measures batch-one autoregressive generation: five prompts x three samples; warm-up excluded; CUDA synchronized; median of three full-workload timings. Numerator counts emitted non-EOS tokens, and elapsed time includes prompt setup/tokenization. EOS can shorten outputs. Trigram is CPU; neural models GPU. Recurrent state is cached; Transformer recomputes context without a KV cache. Throughput characterizes these implementations and sampling settings.

Table B3. Ordinary input-embedding vector norms.

Embedding table	Mean ordinary L2	Std ordinary L2
Random initialization	9.975	0.707
LSTM: random trainable	10.050	0.750
LSTM: frozen Skip-gram	1.481	0.499
LSTM: frozen GloVe	5.204	1.288

Table B3 summarizes vector norms. GloVe covers 1,306,910 of 1,328,463 ordinary-LM-word occurrences in training. PAD and special-row norms are recorded separately in embedding_diagnostics.json. No rescaling was applied. The local parsing cost is a current measurement, not the historical download or pretraining cost.

Appendix C. Matched generation examples

The five prompts are: the movie was; i really enjoyed; the story begins with; the acting was; although the film. The excerpts below use the movie was, sample 0, selected by index for every model/policy. All 180 continuations and token IDs are saved in generation.json. Trigram generation uses (w2,w3)=(0.5,0.4); Table 4 reports diversity and Table B2 reports speed.

Temperature 1; full distribution

Trigram (seed 100000): a goes wages accordingly the performances of the killer snowman cast's to to understand our advancement i'm a performance the from film was to include a good plot must garden stream . this title the <unk> talking the <unk> .

RNN (seed 101000): the credit made a lot prove . those actors have me with a studio flacks , especially the former audience , eating early 90 s , underworld officials so much <unk> with this . plus , maybe <unk> children are

LSTM (seed 102000): directed by joel <unk> as being into the same nonsensical character throughout of the bat ; the irish mother strip - - bettie , poignancy abstract fantastic jokes made the images that gives the influence . the premise is in

Transformer (seed 103000): just busy for television thru out but first , but it was stupid . there are moments of cheap , they put on in life , especially the first place crying wolf . this is a man . fawcett should

Frozen Skip-gram (seed 104000): a very magical , but admittedly the rest of the film still very inept and more simple . although the lyric was simple humanity in view of steady with <unk> or sensationalist in <unk> <unk> , but undermining in this

Frozen GloVe (seed 105000): recorded the same tv show cameo in a slow pace . ' this movie needs forward , the acting more great companion classic . moore was nothing right and the fact that hasn't got a girlfriend - to drawing that

Temperature 1; top-k=40

Trigram (seed 100000): . it starts off the set , the film has received . . .) i mean . and <unk> , a " the man , and in this case both . and it's <unk> <unk> with <unk> <unk> the

RNN (seed 101000): the two stars a lot of its own actors . this is a good movie , especially the most disturbing movie ever made . the <unk> " <unk> " . <unk> character that have just seen it <unk> . the

LSTM (seed 102000): also one of the funniest thing is about the plot that the action is a great story . the story - - that , it's still a lot of the movie that she doesn't have . the acting is in

Transformer (seed 103000): just that one of a <unk> but the characters was great and stupid . the best character i thought that they were on in my life . the first place where the time at the last 5 . it's not

Frozen Skip-gram (seed 104000): a very different , but it's very much better films with his very . i saw my children who have heard the director had even more money and have had been done in <unk> as they . the only way

Frozen GloVe (seed 105000): based on the tv show and that they are no doubt how this movie needs to get the people . it was just a bad script and watching this movie when i saw it on youtube on the box that

Appendix D. Error analysis and paired uncertainty

Table D1. Additional descriptive error groups.

Group	n	LSTM acc.	Transformer acc.	TF-IDF acc.
length<=128	158	67.1%	70.3%	89.9%
length129-256	480	70.2%	71.5%	87.7%
length>256	362	68.5%	69.9%	82.9%
UNK<=2%	375	68.0%	70.9%	89.6%
UNK>2%	625	69.8%	70.6%	84.3%

Table D1 uses token-ID length and per-review UNK share; groups overlap. The broad negation proxy is omitted. For n=158 short reviews, 18 cases favor LSTM and 23 favor Transformer; the 3.16-point difference has exact paired McNemar p=0.533. This post-hoc subgroup check is exploratory.

Table D2 reports macro-F1 uncertainty. The following page examines four random errors and two high-confidence LSTM errors; 0 denotes negative and 1 positive.

Table D2. Classification macro-F1 confidence intervals.

Representation	Macro-F1	95% bootstrap interval
TF-IDF, 50,000d	0.8630	[0.8417, 0.8840]
LSTM + PCA, 100d	0.6698	[0.6398, 0.6989]
LSTM mean, 256d	0.6910	[0.6618, 0.7197]
Transformer mean, 100d	0.7068	[0.6787, 0.7350]

Appendix D (continued). Reading individual errors

The first four cases are the first two saved random errors per LM (seed 4213); the last two are the most confident LSTM errors in each direction. Selected passages use [...] for omissions; full reviews remain in the saved error JSON. Each interpretation below is my reading, not a demonstrated model mechanism or an estimate of error-type prevalence.

LSTM #965: 1 -> 0. "Begotten" is one of the most visually dazzling and mystifying films ever known to man. [...] It's just that everything happens to such a slow pace. [...] Still,it's....I don't know....Glorious...

My reading: The opening excerpt emphasizes praise, but truncating the review hides its final judgment. The full text also criticizes pacing before ending positively; negative plot vocabulary and reservations may distract from that verdict, but this mechanism is untested.

LSTM #922: 0 -> 1. I liked the main characters and the overall story but some scenes are pretty sloppy and confusing. [...] Sound design was a bit poor. I think less would have been best.

My reading: Affection for the characters and story coexists with criticism of execution and sound. I read this as mixed sentiment whose overall negative judgment is missed.

Transformer #987: 0 -> 1. I really liked this quirky movie. [...] It has a realistic edge, with a captivating story line. The main title sequence alone makes this movie fun to watch.

My reading: Even the full review appears positive, despite the recorded negative label. A label inconsistency is plausible but unverified; I retain the supplied label and do not count this as confirmed annotation noise.

Transformer #254: 0 -> 1. A lot of actors have a multitude of good movie roles in their soul. [...] Makes you forget what a terrible movie you're watching. [...] Two stars.

My reading: The positive phrase about movie roles frames a sarcastic, negative review; the explicit verdict is poor. Positive vocabulary alone would misrepresent its sentiment.

LSTM #605: 0 -> 1; predicted-class probability 0.9965. Great premise, poor execution. Cast of great actors is watered down into a poorly written, poorly directed, poorly edited, waste of film.

My reading: Praise for the premise and cast is outweighed by direct criticism of execution. This is a clear negative verdict despite positive words; high classifier confidence does not make it reliable.

LSTM #447: 1 -> 0; predicted-class probability 0.9959. This movie is a masterpiece. Go see it. There's nothing more to say.

My reading: The recommendation and repeated praise in the full text are unambiguously positive to me. Words such as sorrow occur within praise, so isolated negative cues would lose the surrounding meaning; that explanation remains a hypothesis.

References and source notes

[1] Stanford NLP. IMDb dataset card. Dataset identifier: stanfordnlp/imdb.
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https://nlp.stanford.edu/pubs/glove.pdf

[3] Mikolov, T., Chen, K., Corrado, G., and Dean, J. (2013). Efficient Estimation of Word Representations in Vector Space.
https://arxiv.org/abs/1301.3781

[4] Mikolov, T., Sutskever, I., Chen, K., Corrado, G., and Dean, J. (2013). Distributed Representations of Words and Phrases and their Compositionality. NeurIPS.
https://arxiv.org/abs/1310.4546

[5] Hochreiter, S., and Schmidhuber, J. (1997). Long Short-Term Memory. Neural Computation 9(8), 1735-1780.
https://www.bioinf.jku.at/publications/older/2604.pdf

[6] Vaswani, A., et al. (2017). Attention Is All You Need. NeurIPS.
https://arxiv.org/abs/1706.03762

[7] Kingma, D. P., and Ba, J. (2015). Adam: A Method for Stochastic Optimization. ICLR.
https://arxiv.org/abs/1412.6980

[8] Maas, A. L., et al. (2011). Learning Word Vectors for Sentiment Analysis. ACL, pp. 142-150. Dataset split construction: Sections 4.3.1-4.3.2.
https://aclanthology.org/P11-1015/

GloVe data access: https://nlp.stanford.edu/projects/glove/. The vectors are glove.6B.100d.txt from glove.6B.zip. IMDb split construction and HF fields were verified on 15 October 2026.

Artifact provenance

Numerical tables are read directly from saved experiment JSON, feature-based classifier predictions and training histories. The report builder does not train models. Supplementary analysis keeps the original split, labels and vocabulary. The submission code package includes a file manifest; raw external corpora and public embedding files are obtained from their original sources.